

Little Red Dots: Stellar Eruptions, Not Massive Black Holes?

Super-Eddington winds mimic AGN features; engines may be intermediate-mass stars.

arXiv:2606.30711, based on Type IIn SNe and eta Carinae

First author / year / source

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Background

JWST's Little Red Dots show odd spectra unlike known galaxies. Their features resemble stellar eruptions like eta Carinae's Great Eruption.

Are LRDs overmassive black holes or something else? Their true nature is debated.

What this paper is selling

This paper proposes LRDs are super-Eddington stellar engines enshrouded in dense winds. This explains their blackbody-like continua and line profiles without invoking massive black holes.

Key results to remember

- LRDs cluster at ~5000 K like IIn SNe and eta Carinae.
- Wind pseudo-photosphere explains LRD spectra and line shapes.
- Engine mass likely <1e5 solar masses, not overmassive black hole.

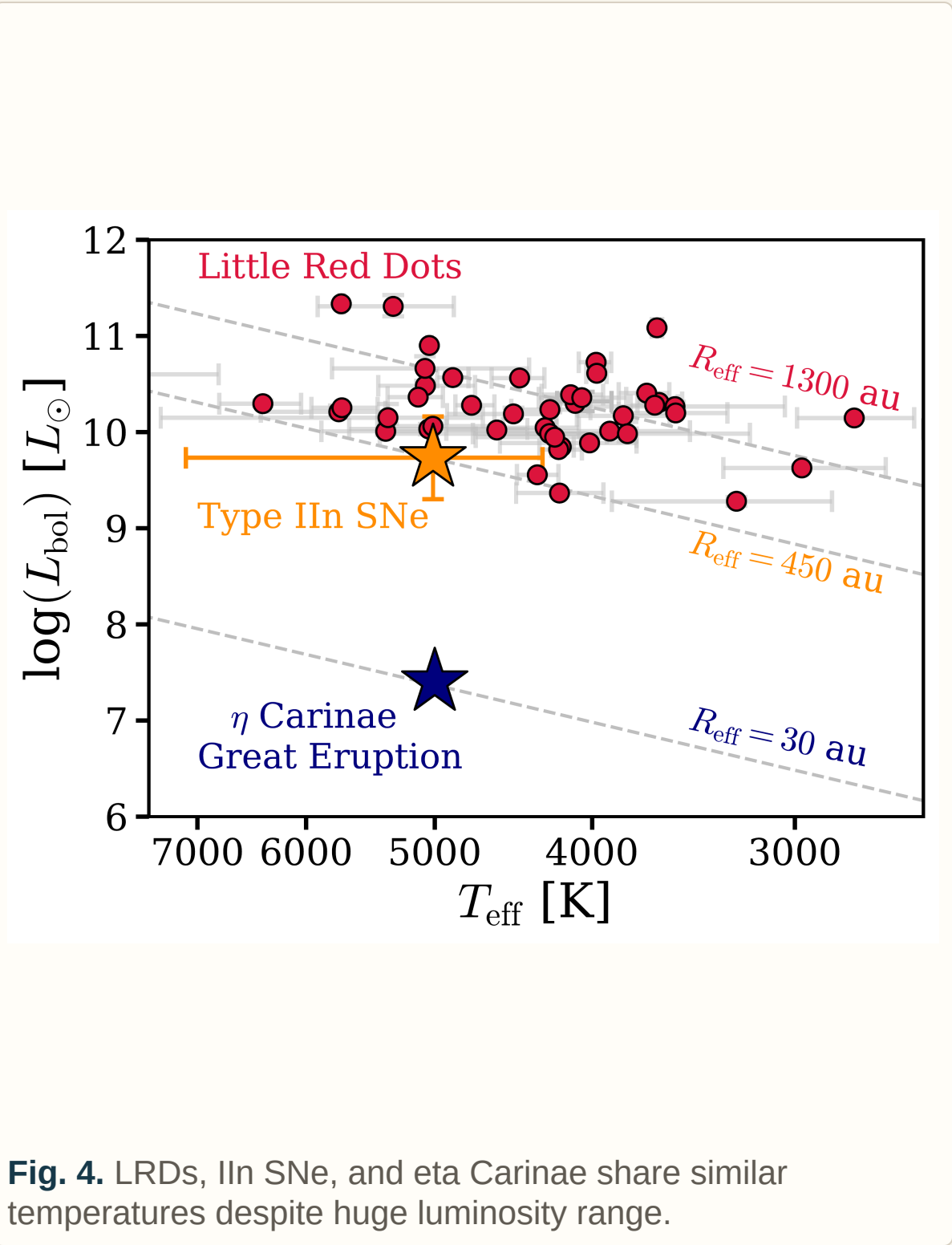


Fig. 4. LRDs, IIn SNe, and eta Carinae share similar temperatures despite huge luminosity range.

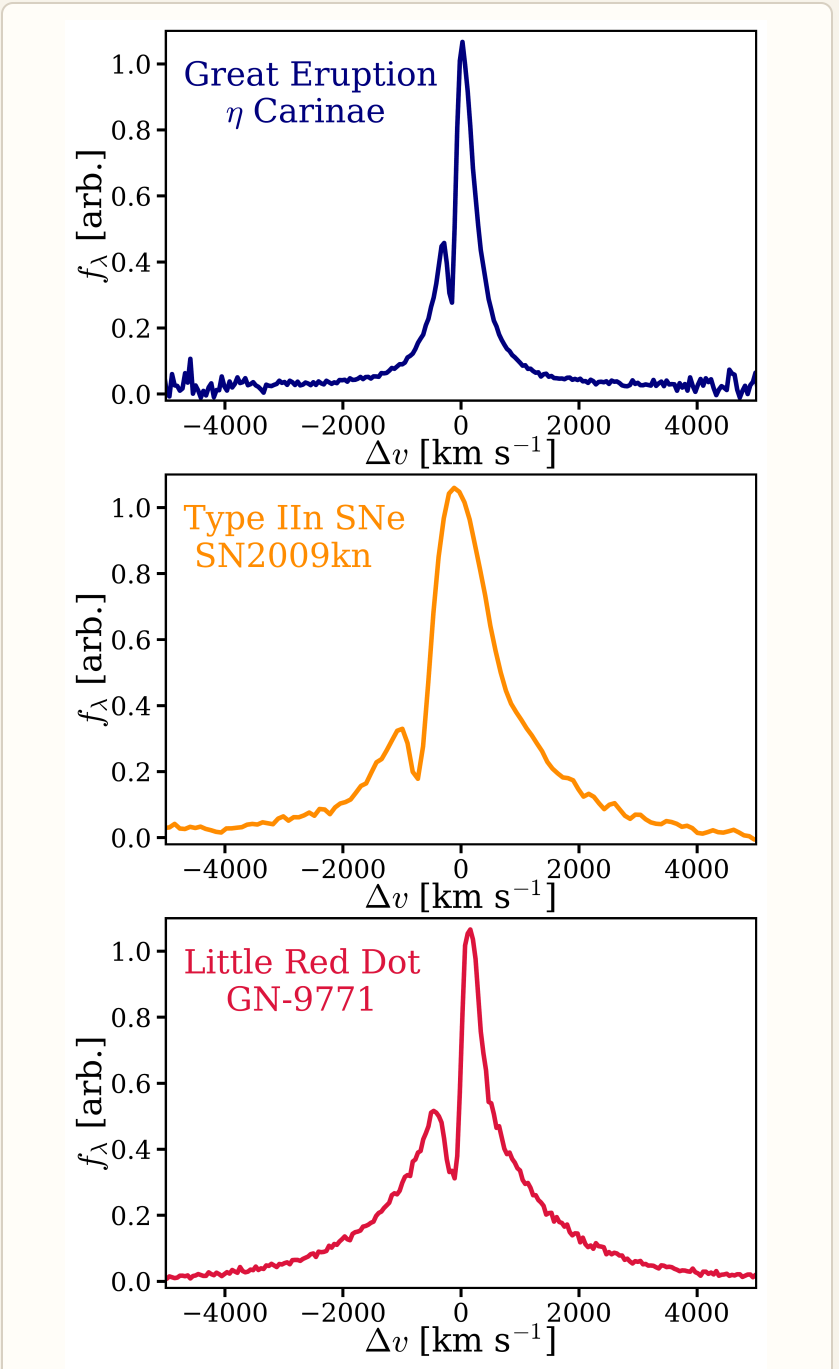


Fig. 1. Halpha profiles from eta Carinae, IIn SNe, and LRDs show striking similarities.